C-MIL-P-18942A (NOrd)
18 December 1956
Superseding
C-MIL-P-18942
30 June 1956

MILITARY SPECIFICATION
PROPELLANT FOR 20MM AIRCRAFT GUN MARK 12

1. SCOPE

1.1 Scope. - This specification covers propellant suitable for use in loading ammunition for the 20 mm Aircraft Gun, Mark 12. As used in this specification, the work "powder" is synonymous with "propellant".

1.2 Type. - The propellant shall be of one type: single base (nitrocellulose).

1.3 Class. - The propellant shall be of one class: Cylindrical, containing metal fouling inhibitor. Although powder is normally furnished in accordance with the formulas, granulations, processes and desired ballistics described herein, powder deviating in these respects may be procured under this military specification when such specific deviations are authorized in the contract or order.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

JAN-D-98	Diphenylamine
JAN-G-155	Graphite (for use in ammunition)
JAN-P-193	Potassium Sulfate (for Ordnance use)
JAN-E-199	Ether, Diethyl
JAN-D-204	Dinitrotoluene (for use in explosives)
JAN-D-218	Dibutylphthalate (for use in explosives)
JAN-N-244	Nitrocellulose (for use in explosives)
JAN-A-463	Alcohol, Ethyl (for Ordnance use)
MIL-M-19719 (NOrd)	Methyl Centralite (symmetrical dimethyl diphenylurea)
JAN-T-458	Tin, Pulverized
MIL-L-18618	Lead Carbonate, Basic, Dry (for Ordnance use)
MIL-P-11960	Propellants, General Specification for Methods of Inspection, Sampling, and Testing
MIL-G-2550	General Specifications for Ammunition except Small Arms Ammunition

STANDARDS

MIL-STD-129	Marking for Shipment and Storage

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DRAWINGS

BUREAU OF ORDNANCE

982521	20 mm Round, HEI
982523	20 mm Round, API
982519	20 mm Round, Target Practice
L. D. 172524	20 mm Gun Mechanism, Mk 12 Mod. Q, List of Drawings, Subassemblies, etc.

ORDNANCE CORPS

D7553247	Action, Test, Caliber .60 and 20 mm, Assembly, List of Parts and Drawings
SK #FSA-6290	Gauge, Pressure, 20 mm, for Gun T118
SK #FSA-6563	Barrel, Test, 20 mm, for Gun T118
C78589	Tool, Obturating Cup, Seating, and Knockout Assembly
A7584596	Obturator
A39430	Piston, Regular
C78160	Gauges, Piston Hole Assembly
B135729	Gauge, Piston Seat (Maximum and Minimum)
A39436	Cylinder, Pressure, Small Arms
49-6-48	Rest, Recoil, Multiple Purpose
FB 22274	Gauge, Air Space, Small Arms Ammunition
B7643660	Reamer, Piston Hole
FD14708	Jig for Drilling Caliber .50, .60, and 20 mm Cartridges, Assembly, and Details
FD16042	Primer Firing Device, Electric, for Ammunition Testing
76-4-46	Box, Packing, Metal liner (Copper), M24, for Smokeless Powders
76-4-56	Box, Packing, Metal-Wood, M17 for Smokeless Powders

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-PM, Volume III, Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by that agency, See 6.4).

2.2 Other Publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulation 49CFR 71-78 - Transportation Interstate Commerce Commission, Explosives and Other Dangerous Articles.

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3. REQUIREMENTS

3.1 Material

3.1.1 Raw Materials. - The raw materials used in the manufacture of propellants purchased under this description shall conform to the specifications listed in 2.1. In addition, the following material shall be of the type, grade or class specified:

3.1.1.1 Nitrocellulose. - Nitrocellulose, Type I or II, grade C, of Specification JAN-N-244, made from cotton cellulose shall be used for small arms propellants. Nitrocellulose, Type I or II, grade C, made from wood cellulose shall not be used unless prior approval of the procuring agency has been obtained. Nitrocellulose recovered from scrap propellant shall not be used without prior approval of the procuring agency.

3.1.1.2 Ethyl Alcohol. - Specification JAN-A-463, grade 2.

3.1.1.3 Graphite. - Specification JAN-G-155, grade III or grade IV.

3.1.1.4 Diethyl Ether. - Specification JAN-E-199, grade A or C.

3.1.1.5 Potassium Sulfate. - Specification JAN-P-193, grade a.

3.1.2 Rework Material

3.1.2.1 Normal Scrap. - Normal scrap colloid from mixers, blocking presses, extruding presses, cutting machines, and other first stage units, if any, prior to solvent recovery of drying shall be collected frequently into covered receptacles, keeping it soft with solvents. This scrap colloid, if clean and uncontaminated, may be used in the mixers, either separately without limit or mixed with new nitrocellulose. It may be further softened, if necessary, in a macerator.

3.1.2.2 Hard Scrap. - Hard colloid or hard rework of the same composition as material in process, from any operation, or loose grain accumulating between solvent recovery and coating operation, if clean and not damaged by heat or sunlight also may be softened with solvents in a macerator and used with new nitrocellulose in mixing. The quantity shall be such as to produce a uniform colloid, but shall be limited to a maximum of 10 per cent of the mixer charge.

3.1.3 Nominal Composition of Propellant. - The propellant shall conform to the following nominal formula in parts by weight, exclusive of coating agents:

Nitrocellulose	100.00
Basic Lead Carbonate added	0.75
Potassium Sulfate added	1.00
Diphenylamine added	1.00

Upon prior approval of the procuring agency pulverized tin may be substituted for basic lead carbonate.

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In order to obtain the required ballistic characteristics, the propellant shall be coated with any of the following deterrents: dinitrotoluene, methyl centralite, dibutyl phthalate, or mixtures of these. The specific deterrent coating to be used will be specified by the procuring agency. The propellant shall be graphite glazed.

3.2 Limits on Added Chemical Constituents. - Limits shall be held on the added chemical constituents of the powder, as specified in Table I in parts by weight, as determined by analysis.

TABLE I - Limits on the Chemical Additives in Parts by Weight, Inclusive of Coating Agent

Additive Chemical Constituent	Maximum	Minimum
Potassium Sulfate	1.00	0.50
Basic Lead Carbonate	1.00	0.60
Diphenylamine	1.00	0.70
Deterrent (Coating Agent)	15.00	5.00
Tin	0.95	0.55

3.3 Form. - Cylindrical grains; single perforation.

3.4 External Moisture. - 1.00 ± 0.25 per cent.

3.5 Total Volatiles. - Maximum 2.35 per cent.

3.6 Residual Solvent. - Maximum 1.10 per cent.

3.7 Hygroscopicity. - (Per cent moisture absorbed at 30°C between exposure conditions of 20 per cent and 90 per cent relative humidity). Maximum 1.75 per cent (See 6.3.2).

3.8 Dust and Foreign Material. - Maximum 0.075 per cent.

3.9 Stability. - When subjected to the 134.5°C stability test, the propellant shall not cause completed fading of the methyl violet test paper to a salmon pink color in less than 25 minutes. Explosion shall not occur in less than 5 hours.

3.10 Loading Characteristics. - The propellant shall load satisfactorily in all loading machines of the type currently in use for the appropriate ammunition.

3.11 Bulk Density. - The bulk density of the propellant shall be such that linear compression of the propellant charge established to develop the required velocity shall not exceed 0.10 inch when the propellant charge is machine loaded and the appropriate bullet is seated to proper depth in the charged case (see 6.3.2.)

3.11.1 Weight Variation. - The extreme variation in weight among 50 consecutive charges of propellant thrown from a loading machine of appropriate type, using the determined weight of charge established in the velocity firing, shall not be more than 10 grains.

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3.12 Function

3.12.1 Ejection Time (Action Time) and Firing Time Interval. - (See 6.3.2) When assembled in the 20 mm target practice round, the propellant shall ignite properly with the standard primer. Ejection time (action time), defined as the interval between closing the firing key and the emergence of the projectile from the muzzle and the firing time interval, defined as the difference between the shortest action time and longest action time of fifty rounds as in burst fire exclusive of the first round, shall be as specified below:

Mean Ejection Time	.004 sec max.
Standard Deviation of Ejection Time	.0005 sec max.
Firing Time Interval	.0023 sec max.

3.12.2 Fouling. - (See 6.3.2) The propellant shall not produce excessive fouling of the action, barrel, muzzle attachments, or gas cylinders of automatic arms that would impair the accuracy of the ammunition or prevent the normal function of the arm. Following the firing as prescribed in 4.4.3.3, the barrel shall show no corrosion or fouling after being cleaned by the ordinary method with bristle brushes and patches.

3.13 Ballistic Requirements

3.13.1 Propellants shall, unless otherwise specified, meet the following ballistic requirements when assembled in 20 mm round, target practice, using calibration components:

Average Corrected Velocity (Burst Velocity)	3290 to 3350 f/s
Standard Deviation of Burst Velocity	30 f/s max.
Maximum Mean Pressure (Mann Barrel)	58000 psi (copper)
Standard Deviation of Pressure	2500 psi (copper) max.

3.13.2 The following is provided for informational purposes only. When specified for a particular cartridge, propellants shall meet the basic ballistic requirements for the cartridge as provided in the applicable ammunition acceptance specification and as specified in Table II.

TABLE II - Ballistic Requirements

20 mm Round	Muzzle Velocity (Measured full automatic) f/s	Maximum Standard Deviation f/s	Maximum Mean Pressure (Mann Barrel) psi
Practice Dr. 982519	3290 to 3350	40	58000
API Dr. 982523	3290 to 3350	40	58000
HEI Dr. 982521	3290 to 3350	40	58000

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3.14 Process. - The process of manufacture used by the manufacturer shall be approved by the procuring agency in writing before the manufacturer proceeds with production. In case any change in the process of manufacture appears to be questionable, the contracting officer shall bring the matter to the attention of the Chief of the Bureau of Ordnance or the Chief of Ordnance and notify the contractor of this action in writing within 24 hours.

3.14.1 Description Sheets. - With every lot of propellant submitted for acceptance, the manufacturer shall furnish on official blanks eight copies of a description sheet giving a complete history of its manufacture.

3.14.2 Dehydration of Nitrocellulose. - Nitrocellulose shall be dehydrated with ethyl alcohol. At least 1.10 pounds of alcohol per pound of dry nitrocellulose in the press charge shall be pumped through the block, including the residual alcohol left in the block. Excess alcohol shall be removed by pressure leaving no more than is required for mixing.

3.14.3 Mixing. - Using suitable mechanical mixers, the blocks of dehydrated nitrocellulose shall be mixed with the required amount of solvent and other ingredients. The amount of solvent used is left to the discretion of the manufacturer, but solvents and mixing time shall be sufficient to give thorough colloiding and distribution of all ingredients. Further colloiding, if necessary, shall be done in a macerator.

3.14.4 Pressing. - The charge from the mixer or macerator shall be blocked for sufficient time and at a pressure previously determined to give thorough consolidation. The colloid shall then be blocked and pressed through screens and dies to form strands which are granulated by passing through cutting machines into grains of such dimensions that, after proper drying, will have the final dimensions and perforations desired. The screens used shall be such as to reduce the amount of uncolloided nitrocellulose in the finished grain to a minimum. Material left on these screens shall not be reworked unless found free of foreign particles.

3.14.5 Drying. - The propellant shall be subjected to a solvent recovery process at a temperature not exceeding 60°C. If water drying is used, the temperature shall not exceed 55°C. If air drying is used, the temperature shall not exceed 55°C. The rate at which the temperature is raised to these limits is left to the discretion of the manufacturer, but the rate must be such as to prevent distortion of the propellant grains.

3.14.6 Coating. - The application of an approved compound to the grains of propellant in quantities necessary to give the required ballistics shall be performed under the conditions as prescribed by the contractor.

3.14.7 Glazing. - The propellant shall be glazed with graphite in such a manner and for such a period of time as to give uniform, complete coverage of the individual pellets.

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3.14.8 Sieving and Cleaning. - The propellant shall be sieved and cleaned to remove doublets, clusters, tailings, short cuts, splinters, chips, dust, loose graphite and foreign materials.

3.14.9 Blending. - The propellant shall be blended uniformly in lots of prescribed size. Prior to or during the blending operation, moisture may be added, if necessary, in such a way as to insure its uniform distribution and in such quantity as to meet the requirements for moisture content of the finished powder specified in 3.4.

3.14.10 Finish. - The propellant shall have a clean, polished appearance and shall be uniform in granulation, with surface characteristics such that it will flow and charge uniformly when machine loaded.

3.15 Workmanship. - The finished propellant shall be a homogeneous colloid, as free as possible from uncolloided nitrocellulose. Material going through processes shall be kept in covered containers when required, and also protected from direct sunlight, dirt, acids, or contamination. Material found to be harmfully discolored or contaminated shall be removed from process and destroyed. Contaminated material will include any propellant which has been made from a wrong formula or which is found to contain material not listed in specifications or not otherwise specifically authorized.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Lot. - Unless otherwise specified, the lot size shall be between 25,000 and 60,000 pounds and shall consist of homogeneous material manufactured under essentially the same process. If a lot of propellant is retreated or redried, it shall constitute a new lot. The lot number shall be affixed with the letter "A" to indicate retreatment, and the lot shall be subjected to a complete acceptance test.

4.2 Sampling. - The Government inspector shall select six boxes at random from a low number through to a high number. From each selected box the quantities of propellant specified in Table III shall be taken.

TABLE III - Sample Quantities

Chemical	Ballistic	Surveillance	Loading
1 lb	5 lb	1 lb	3 lb

Sampling shall be done with care not to handle the propellant grains with the hands or allow the open boxes to become contaminated. Samples for all four categories of Table III shall be prepared by combining the material from the six boxes. In each case, the container for the composite should be labeled with the correct sample type indicated in Table III. The "chemical" and "ballistic" samples should be sealed with a Government seal. The surveillance samples shall be shipped to the Naval Powder Factory, Indian Head, Maryland, for Navy material and to other designated points as required for Army material. The loading sample shall be used for testing, machine loading, ejection time (hang-fire), and fouling characteristics. When

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shipment of powder samples is made to a different facility by common carrier, preparation for shipment shall be in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles.

4.3 Inspection. - Unless otherwise specified inspection shall be made at the place of manufacture and shall be in accordance with Specification MIL-G-2550. Propellant which cannot be tested at the source shall be forwarded to such facility for the desired tests as may be designated by the procuring agency. The official samples described in 4.2 shall be taken promptly after the submission of the lot for acceptance.

4.4 Tests

4.4.1 Chemical Composition and Physical Tests. - Tests shall be conducted in accordance with applicable test methods given in MIL-P-11960 to determine compliance with the requirements of 3.1.3, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8 and 3.9. Tests for external moisture (3.4) are conducted in accordance with method 102.1 of MIL-P-11960B. Tests for total volatiles (3.5) are conducted in accordance with method 103.3 of the above MIL-Specification. Other procedures may also be used if mutually approved by the procuring agency and the contractor. (See 6.3.2 and 6.3.3.)

4.4.2 Loading Characteristics

4.4.2.1 Linear Compression of Propellant Charge. - Determine the linear compression of the propellant charge by machine loading ten cartridge cases, using calibration cases, with the weight of charge established in the propellant assessment (4.4.3.4.5). Do not insert the projectile. Measure the linear distance (either positive or negative, as may occur) with an airspace gage, as shown on Drawing FB22274, between the top of the propellant column and the mouth of the cartridge case. Subtract the seating depth of the bullet from this measurement to determine the airspace. A negative reading indicates compression, and shall not exceed the limit specified in 3.11.

4.4.2.2 Weight Variation. - The determined weight of charge shall be loaded consecutively at normal rhythm of operation, from mechanically actuated charging blocks of appropriate design, into fifty (50) cases of the type for which the propellant is intended. Each charge so dropped shall be verified by weighing the propellant charge on a balance sensitive to ± 1 grain. Care shall be exercised to prevent propellant from losing or gaining moisture.

4.4.3 Function and Ballistics

4.4.3.1 Assembled weapons used in acceptance of propellants shall have the following characteristics:

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	Breeching* Space	Lock Clearance	Crush-up
Gun, Automatic 20 mm, Mk 12, Mod 0	5.118 + .007 in. - .005 in.	.010 in. max.	0.037 + .005 in. - .019 in.
Action Test, Electric, Cal. .60 and 20 mm			0.003 to .010 in.

* As defined on page 3 of Section OPM 7-11 of ORD-M-608-PM Volume III.

4.4.3.2 Ejection Time (Action Time) and Firing Time Interval

4.4.3.2.1 Fifty 20 mm target practice rounds shall be assembled as described in 4.4.3.4.1. These rounds shall be burst fired in the 20 mm automatic gun, Mark 12, and ejection times (action times) recorded.

4.4.3.3 Fouling. - After fully automatic firing of 25 target practice rounds, utilizing ordinary (not calibration) components, the weapon shall be visually inspected for fouling.

4.4.3.4.1 Ballistics

4.4.3.4.1.1 Components and Assembly. - All firings shall be made with rounds utilizing calibration components. Rounds shall be rubber crimped to have a pull-out force between 1300 and 1700 pounds. Calibration components are defined as follows:

Projectiles: The weight shall be 1700 ± 5 grains
Projectiles shall be 100% inspected for dimensional tolerances and shall fall within the tolerances of BuOrd Drawing No. 982323, and shall be from one lot.

Cases: Cases shall be 100% inspected for dimensional tolerances and shall fall within the tolerances of BuOrd Drawing No. 517534, and shall be from one lot.

Primers: Primers shall be from a previously accepted primer lot containing approximately 50,000 primers.

Cases used in Mann Barrel firings shall be modified as follows:

The cases of both the test ammunition and the reference ammunition shall have been drilled with a NO. 19 (0.166 inch) drill in a position such that the drilled hole can be centered under the piston hole, and the drilled case shall have been wrapped with one turn of cellulose tape, so that the hole is centrally positioned under the tape, prior to assembly. Initially uncompressed copper pressure cylinders shall be used.

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4.4.3.4.2 Range. - All propellant acceptance firings shall be conducted on a range which has been equipped with instruments and accessories of proven acceptability which have been previously checked and are known to be in satisfactory working condition. The temperature of the propellant and ammunition components shall be 70°F ± 2°F at the time of assembly and firing.

4.4.3.4.3 Weapons. - Firing for velocity and pressure will require the use of barrels which, when converted to accept pressure gauges, and assembled to action, test, electric, caliber .60 and 20 mm (Drawing D7553247) and fired with reference powder* (hand assembled at the assessed weight of charge with appropriate calibration components) gives an average velocity not deviating from the assessed ballistic level of the reference powder by more than ± 35 feet per second or ± 3500 pounds per square inch in pressure. The burst velocity tolerance for machine gun barrels shall be ± 50 f/s. Physical characteristics of the assembled weapons are as stated in 4.4.3.1. Weapons chosen for production powder work shall not be used for experimental powders of nonstandard composition.

4.4.3.4.4 Propellant Blend. - The composite ballistic sample described in 4.2 shall be used to establish the weight of powder charge required to develop the specified velocity and to hand assemble cartridges with the established weight of charge for acceptance firing to obtain the average corrected velocity and the average corrected pressure.

4.4.3.4.5 Procedure (see 6.3.5)

4.4.3.4.5.1 Master Standard Powder* Assessment. - Charge-velocity-pressure relationships of the powder lot to be designated a master powder shall be established in Mann Barrels and Machine Gun Barrels as follows:

The master powder charge weight assessment shall be determined from firings conducted in five new machine gun barrels in each of three different days. Each firing in each barrel on each day shall consist of one fifteen-round burst, at each of three charge levels. One charge level shall be that expected to produce approximately service velocity as estimated from probing rounds. The other two charges shall be approximately 15 grains above and below this level. The first round of each fifteen-round group shall be considered a conditioning round and shall be disregarded in computing average velocities. The grand means of the velocities of the three charge levels fired in the five guns on the three days shall be found and a least squares curve fitted to these means. The master standard charge weight assessment shall be the charge weight determined from this curve which should produce a burst velocity of 3320 f/s in an average new gun.

The master powder pressure assessment shall be determined from firings conducted in five new Mann Barrels on each of three different days. Each firing in each barrel on each day shall consist of three eleven-round groups, one group at each of three charge levels. One charge level shall be approximately the master charge assessment determined above. The other two levels shall be approximately

* "Master Standard Powder" is synonymous with "Master Powder", "reference powder", or "reference propellant", as used in this Specification.

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15 grains above and below this level. The first round of each eleven-round group shall be considered a conditioning round and shall be disregarded in computing average pressures. The grand means of the pressures of the three charge levels fired in the five guns on the three days shall be found and a curve fitted to these means. The master powder pressure assessment shall be the pressure determined from the curve which is produced in the Mann Barrel by the master powder assessed charge weight. This pressure must not exceed 58,000 psi copper pressure.

Master powder barrels shall be selected with great care. Barrels shall be chosen whose dimensions are close to nominal drawing dimensions. The barrels shall have been broken in by firing a sufficient number of cutting rounds (to eliminate erratic velocities caused by wearing of the barrel) before firing the master powder. If during the master powder firing, any individual barrel produces ballistics that are significantly different at the .05 level of confidence from the other barrels, the results from that barrel shall be disregarded. A new barrel shall be selected and ballistics obtained.

Barrels used for master powder assessment shall be kept in reserve for any future master powder firings.

4.4.3.4.5.2 Test Powder Assessment. - Charge-velocity-pressure relationships of a test powder shall be determined by the matched powder method as follows:

The charge weight assessment shall be determined from firings conducted in two machine gun barrels on one day. Each firing in each barrel shall consist of one fifteen-round burst of the test powder at each of three charge levels, and one fifteen-round burst of the master powder. One charge level of the test powder shall be that expected to produce approximately service velocity (estimated from probing rounds if necessary) and the other two charges shall be approximately 15 grains above and below this level. The master powder charge shall be the master powder charge weight assessment as determined by the procedure of (4.4.3.4.5.1). The first round of each fifteen-round group shall be considered a conditioning round and shall be disregarded in computing average velocities. The means of the velocities of the three charge levels of the test powder fired in each of the two barrels shall be found and least squares curves fitted to the two sets of means. From each curve the charge weight shall be selected which should produce a burst velocity equal to the mean velocity of the day's firing of the master powder. The test powder charge weight assessment shall be the average of these two charge weights, unless the two charge weights differ by more than five grains. If this latter situation occurs, additional firings shall be conducted in at least one other gun barrel, and the assessed charge weight determined from a statistical weighing of three or more gun firings.

The pressure assessment shall be determined from firings conducted in one Mann Barrel on one day. This firing shall consist of one eleven-round group of the test powder at each of three charge levels, and one eleven-round group of the master powder. One charge level of the test powder shall be approximately the assessed charge weight determined above, and the other two charges shall be approximately 15 grains above and below this level. The master powder charge shall be the master powder charge weight assessment as determined by the procedure of (4.4.3.4.5.1). The first round of each eleven-round group shall be considered a

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conditioning round and shall be disregarded in computing average pressures. The means of the pressures for the three charge levels of the test powder shall be found and a curve fitted to the means. The pressure that corresponds to the assessed charge weight of the test powder shall be determined from this curve and this pressure corrected by adding algebraically the difference between the mean pressure produced by the master powder in the day's firing and the assessed pressure of the master powder. This corrected pressure shall be the assessed pressure of the test powder. This pressure must not exceed 58,000 psi copper pressure for an acceptable powder.

Velocity curves shall be considered to be linear; the over-service charge pressure shall be assumed not to fall below the linear extrapolation through the below-service and near-service charge pressures.

4.5 Retest and Rejection. - Should a lot of powder fail to pass one or more of the tests specified, the inspector shall notify the manufacturer, and upon the request of the manufacturer shall repeat the failing test or tests. At the time the repeated test or tests are made, the same test or tests shall be made with the master powder and the results obtained shall be used as the basis of comparison for judging the characteristics of the lot of powder. Should the master powder fail to give results complying with the specification, the Government Inspector shall investigate the conditions of testing, as well as the master powder in order to determine and to remove the cause of failure before proceeding with the repeated test or tests of the lot of powder for acceptance. If a lot of powder fails to pass the required retest or retests, as described above, it shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing. - The powder shall be packed in accordance with one level as follows:

5.1.1 Level B. - The powder shall be packed in clean containers complying with the applicable drawing specified in 2.1, or as otherwise approved by the procuring agency. All containers shall be tested with compressed air for air tightness by the manufacturer. Containers conforming to Drawing 76-4-46 shall withstand an internal pressure of 1/2 to 1 psi for 1 minute without leakage.

5.2 Marking. - Marking to insure safe handling shall be in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles. Each container shall be marked as prescribed in MIL-STD-129 with additional marking indicating purchase order or contract number, place and date packed, lot number, name, type and class of powder, and the particular ammunition for which the contained powder is intended.

6. NOTES

6.1 Ordering Data. - Invitation for bids and contracts or orders should specify the title, number, and date of this military Specification.

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6.2 Although powder is normally furnished in accordance with the formulas, granulations, processes, and desired ballistics described herein, powder deviating in these respects may be procured under this military Specification when such specific deviations are authorized in the contract or order.

6.3 Miscellaneous Notes

6.3.1 Surveillance. - Surveillance tests at 65.5°C on Navy propellants will be conducted at the Naval Powder Factory, Indian Head, Maryland, for information purposes.

6.3.2 After a powder of the type desired has been initially evaluated and found to be acceptable, usually after the first five lots have been produced consecutively at the facility involved, for purposes of continuing procurement, the tests for compliance with the requirements of 3.7, 3.11, 3.11.1, 3.12.1, 3.12.2 or any of them, may be waived upon authorization of the procuring agency.

6.3.3 At the time the chemical and ballistic samples are taken by the Government Inspector, the manufacturer may, upon authority of the contracting officer, take for his own use samples of the powder in similar quantities and in like manner. These samples may be used by the manufacturer for making ballistic or chemical tests, or both, for his own information.

6.3.4 Copper crusher gauges other than those specified, or other pressure measuring systems may be used if approved by the Chief of the Bureau of Ordnance or the Chief of Ordnance. Corresponding changes in the pressure limits would then be specified, if required.

6.3.5 Copies of assessments for 20 mm reference powder may be obtained from the Chief of the Bureau of Ordnance, Department of the Navy, Washington 25, D. C. or Chief of Ordnance, Department of the Army, Washington 25, D. C.

6.4 Drawings required in this specification that are required for Navy procurement functions should be requested from the Superintendent, U. S. Naval Gun Factory, Washington 25, D. C.

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